📄 Bash Complete Introduction Guide

A beginner-friendly guide to command-line scripting and automation.

📘 What is Bash?

Bash (Bourne Again Shell) is a **command-line shell and scripting language** used in Linux and macOS.

It allows you to:

- Run commands ⚡
- Automate tasks 🔁
- Manage systems 💻

In simple terms:

Bash = Control your computer using commands

🕰 History of Bash

Year	Event
1989	Bash released
Creator	Brian Fox
Developed For	GNU Project

Bash replaced older Unix shells and became widely used.

⭐ Why Use Bash?

Key Benefits

✓ Automate repetitive tasks
✓ Fast and efficient
✓ Powerful system control
✓ Widely used in DevOps
✓ Available on most systems

⚡ Core Features of Bash

1. Command Execution

Run system commands easily.

2. Scripting

Write scripts to automate tasks.

3. Variables

Store and reuse values.

4. Loops & Conditions

Control flow in scripts.

5. File Management

Handle files and directories.

🛠 Running Bash

Open Terminal:

- Linux 🐧
- macOS 🍎
- Windows (Git Bash / WSL)

💡 Basic Commands

```
pwd        # Show current directory
ls         # List files
cd folder  # Change directory
mkdir test # Create folder
rm file    # Delete file
```

📦 Variables

```
name="John"
echo $name
```

🔧 User Input

```
read name
echo "Hello $name"
```

🔁 If Condition

```
if [ $age -ge 18 ]
then
  echo "Adult"
else
  echo "Minor"
fi
```

🔄 Loops

For Loop

```
for i in 1 2 3
do
  echo $i
done
```

While Loop

```
count=1
while [ $count -le 5 ]
```

```
do
  echo $count
  ((count++))
done
```

📂 File Handling

```
touch file.txt
cp file.txt backup.txt
mv file.txt newfile.txt
rm file.txt
```

🎁 Functions

```
function greet() {
  echo "Hello $1"
}

greet John
```

⚛ Bash in Real Life

Used in:

- DevOps ⚙
- System administration 🖥
- Automation scripts 🔁
- Server management 🌐

□ Example Script

```
#!/bin/bash

echo "Enter your name:"
read name
echo "Hello $name"
```

Run:

```
bash script.sh
```

□ Advantages of Bash

✓ Fast and lightweight
✓ Great for automation
✓ Pre-installed on Linux/macOS
✓ Powerful system access

● Disadvantages of Bash

• Syntax can be tricky
• Not ideal for large applications
• Limited debugging tools

🏆 Best Practices

✓ Use meaningful variable names
✓ Comment your scripts
✓ Handle errors properly
✓ Keep scripts simple
✓ Test before running

📌 Quick Summary

Bash helps you:

- Automate tasks
- Manage systems
- Work efficiently with terminals

✨ Tips to Make Your PDF Look Professional

• Use dark terminal theme ❤️
• Highlight commands
• Add icons and sections
• Use fonts like:

- Source Code Pro
- JetBrains Mono
- Roboto

Recommended Layout:

- Cover Page
- Table of Contents
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